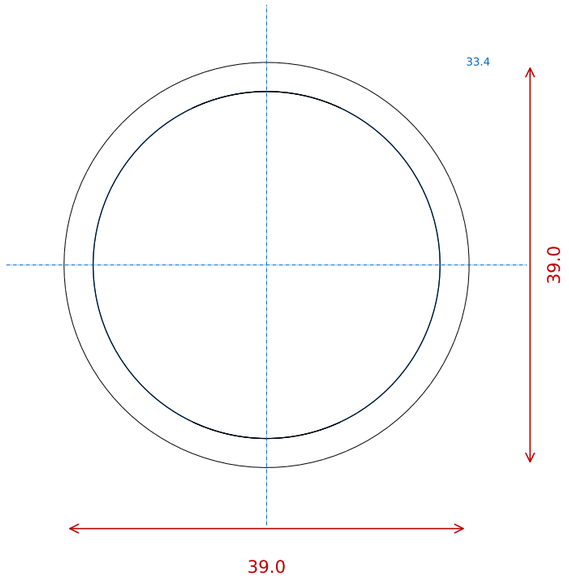
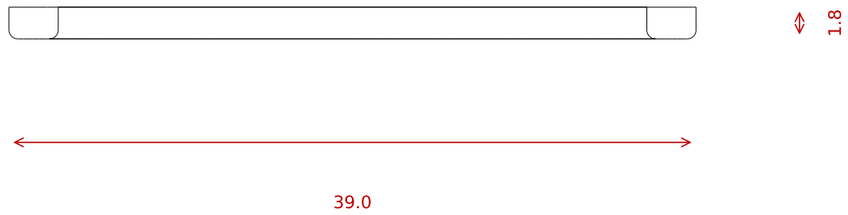
Front (XZ)



Top (XY)



Right (YZ)



Overall size X x Y x Z (mm)

39.0 x 1.8 x 39.0

Volume 0.6 cm3

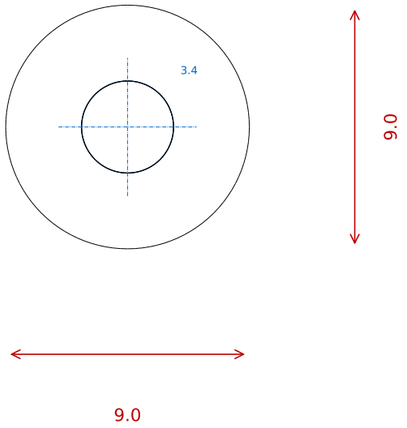
Print: 0.7 g, 0.12 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 33.4/y/1

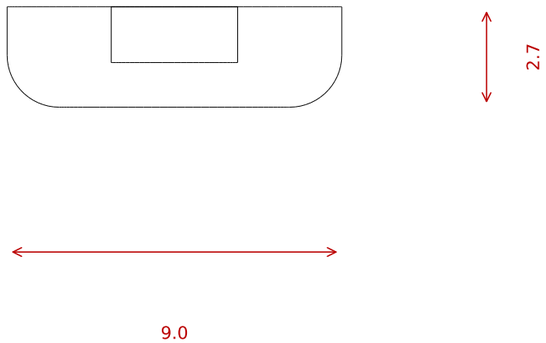
General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Display bezel	Team MEDTHAI	Scale: fit	Sheet 1/9
Spiro - DIY IoT spirometer (v8) file: bezel.stl	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A

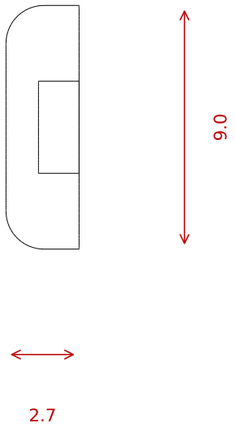
Front (XZ)



Top (XY)



Right (YZ)



Overall size X x Y x Z (mm)

9.0 x 2.7 x 9.0

Volume 0.1 cm3

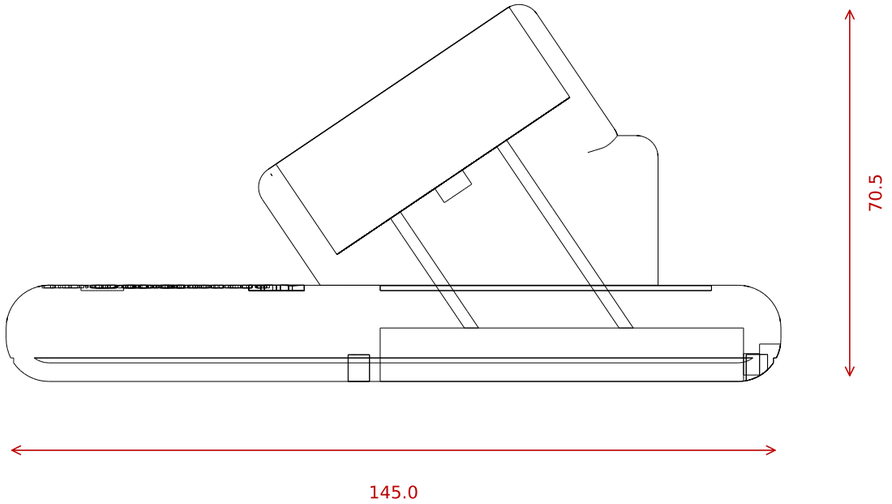
Print: 0.2 g, 0.09 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 3.4/y/1

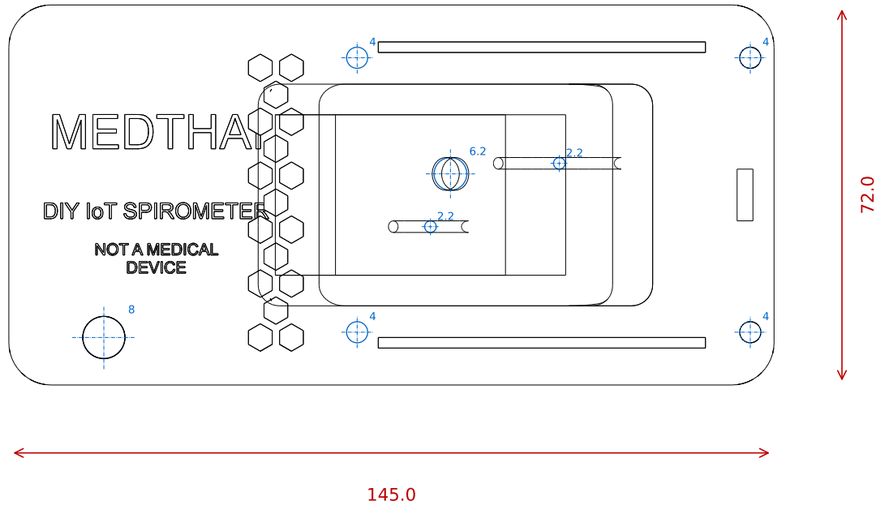
General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Button cap	Team MEDTHAI	Scale: fit	Sheet 2/9
Spiro - DIY IoT spirometer (v8) file: button_cap.stl	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A

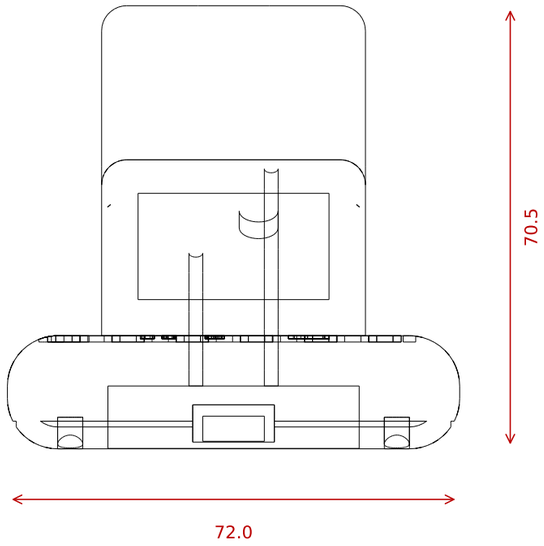
Front (XZ)



Top (XY)



Right (YZ)



Overall size X x Y x Z (mm)

145.0 x 72.0 x 70.5

Volume 225.6 cm3

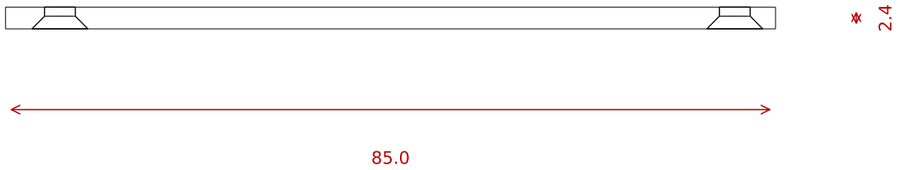
Print: 137.2 g, 7.7 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 2.2/z/2; 4/z/4; 6.2/z/1; 8/z/1

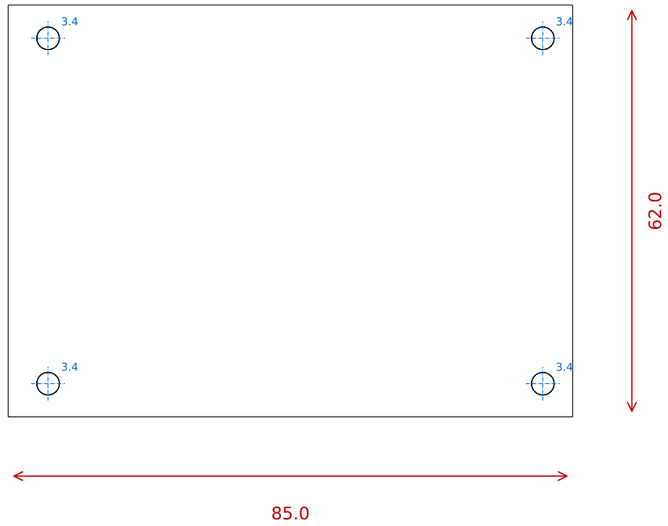
General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Charging dock Spiro - DIY IoT spirometer (v8) file: dock.stl	Team MEDTHAI	Scale: fit	Sheet 3/9
	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A

Front (XZ)



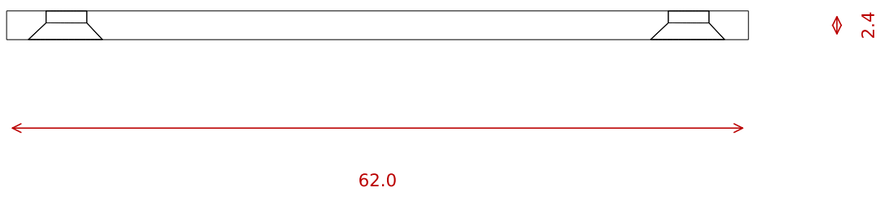
Top (XY)



Overall size X x Y x Z (mm)

85.0 x 62.0 x 2.4

Right (YZ)



Volume 12.5 cm3

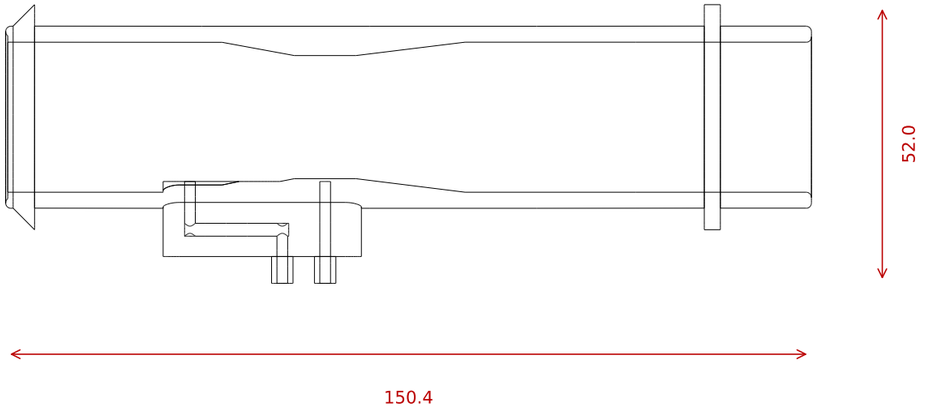
Print: 15.9 g, 0.96 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 3.4/z/4

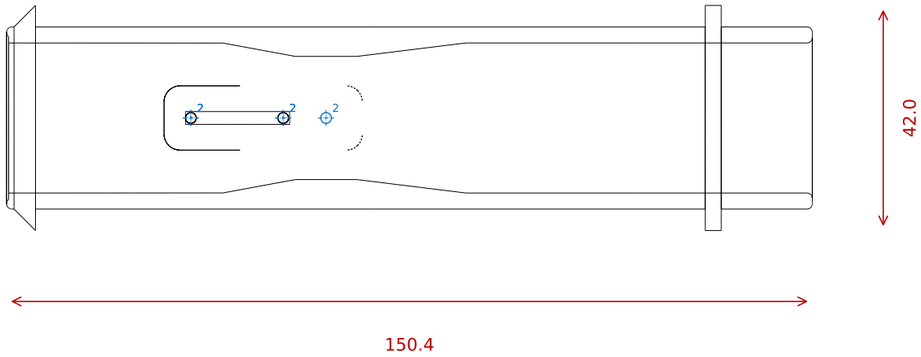
General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Charging dock, base plate	Team MEDTHAI	Scale: fit	Sheet 4/9
Spiro - DIY IoT spirometer (v8) file: dock_bottom.stl	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A

Front (XZ)



Top (XY)



Right (YZ)

Overall size X x Y x Z (mm)

150.4 x 42.0 x 52.0

Volume 56.2 cm3

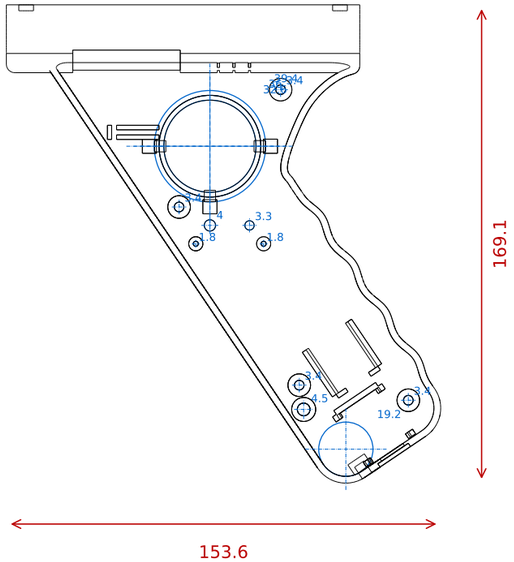
Print: 61.9 g, 3.52 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 2/z/5; 2.4/x/1; 23/x/1; 28/x/2

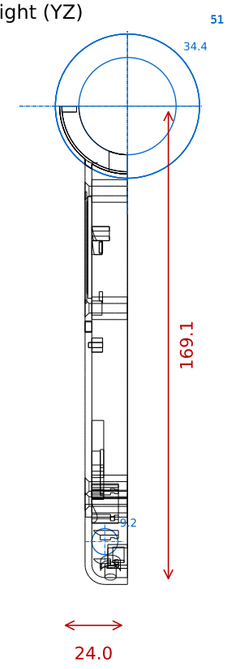
General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Venturi flow tube D28/d23 Spiro - DIY IoT spirometer (v8) file: flow_tube.stl	Team MEDTHAI	Scale: fit	Sheet 5/9
	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A

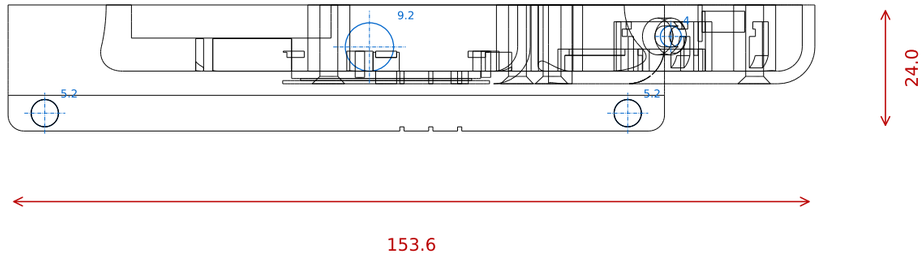
Front (XZ)



Right (YZ)



Top (XY)



Overall size X x Y x Z (mm)

153.6 x 24.0 x 169.1

Volume 58.2 cm3

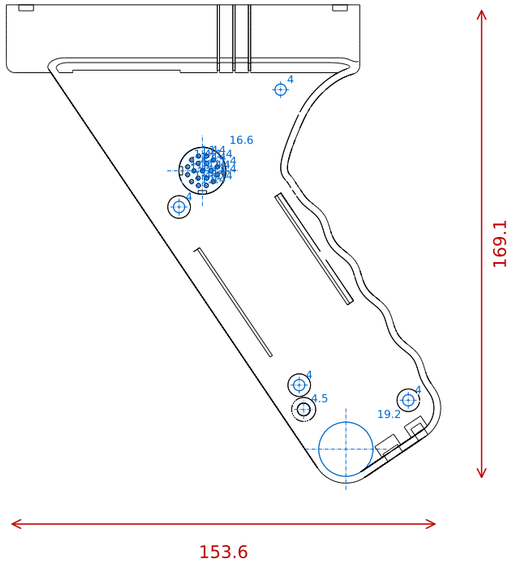
Print: 71.6 g, 4.06 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 1.8/y/2; 3.3/y/1; 3.4/y/4; 4/y/1; 4/z/1; 4.5/y/1; 5.2/z/2; 9.2/x/1; 9.2/z/1; 19.2/y/1; 32.6/y/1; 34.4/x/1; 36/y/1; 39.4/y/1; 51/x/3

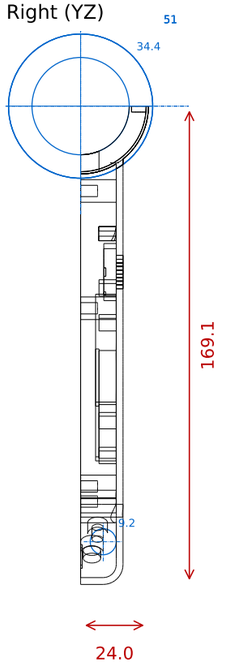
General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Housing, left half	Team MEDTHAI	Scale: fit	Sheet 6/9
Spiro - DIY IoT spirometer (v8) file: housing_left.stl	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A

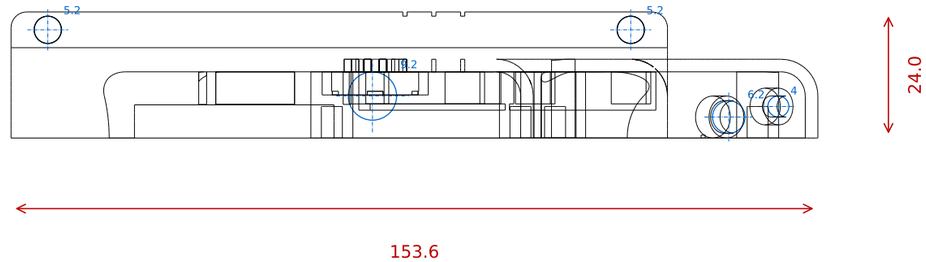
Front (XZ)



Right (YZ)



Top (XY)



Overall size X x Y x Z (mm)

153.6 x 24.0 x 169.1

Volume 62.0 cm3

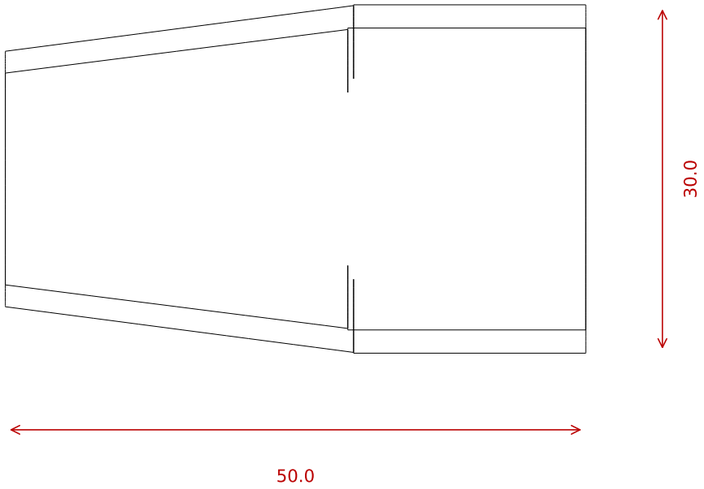
Print: 76.2 g, 4.31 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 1.4/y/19; 4/y/4; 4/z/1; 4.5/y/1; 5.2/z/2; 6.2/z/1; 9.2/x/1; 9.2/z/1; 16.6/y/1; 19.2/y/1; 34.4/x/1; 51/x/3

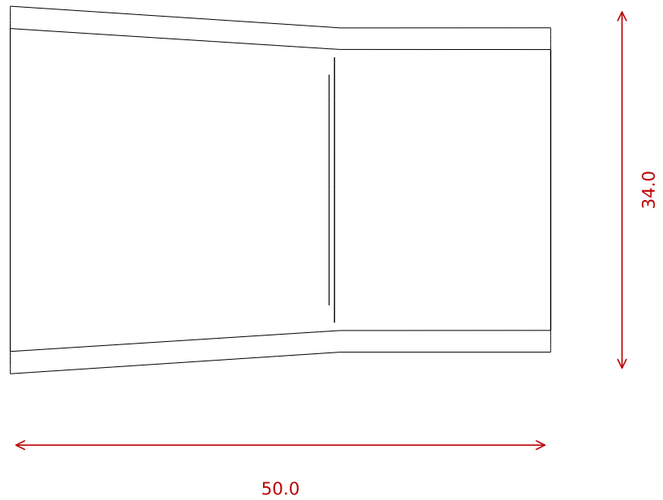
General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Housing, right half	Team MEDTHAI	Scale: fit	Sheet 7/9
Spiro - DIY IoT spirometer (v8) file: housing_right.stl	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A

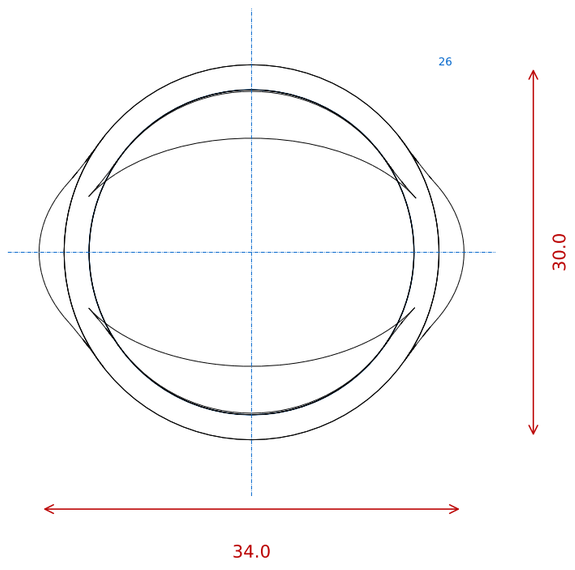
Front (XZ)



Top (XY)



Right (YZ)



Overall size X x Y x Z (mm)

50.0 x 34.0 x 30.0

Volume 8.5 cm3

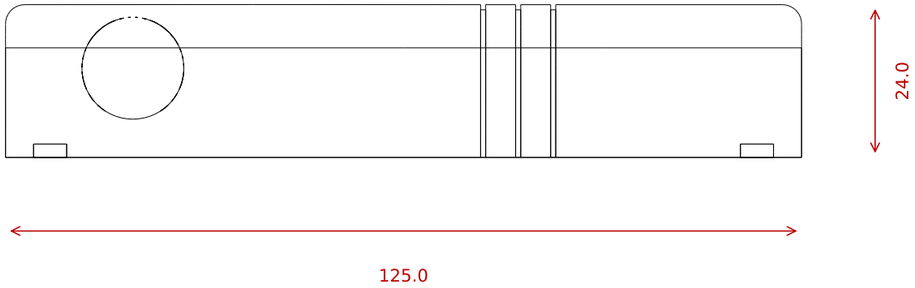
Print: 10.9 g, 0.68 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 26/x/1

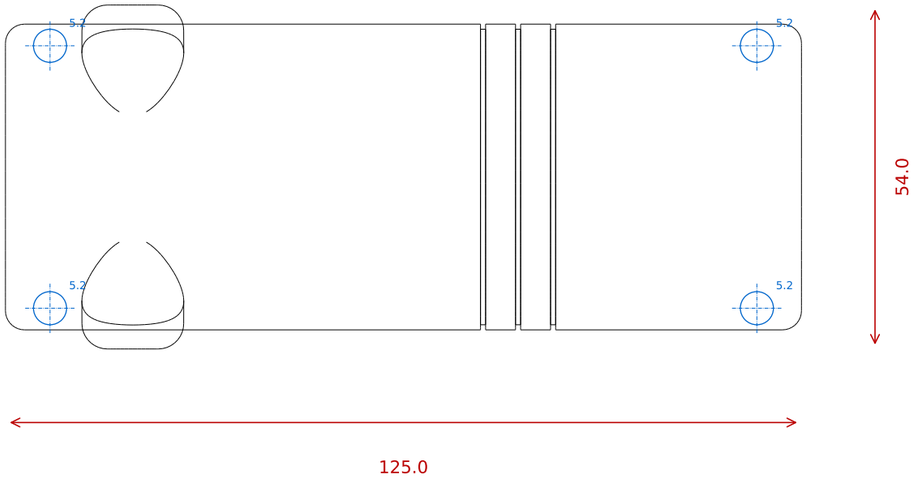
General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Mouthpiece (single use)	Team MEDTHAI	Scale: fit	Sheet 8/9
Spiro - DIY IoT spirometer (v8) file: mouthpiece.stl	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A

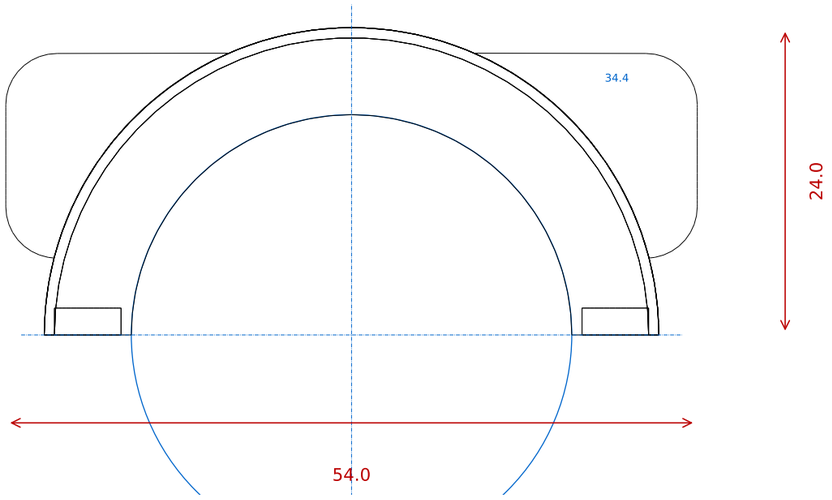
Front (XZ)



Top (XY)



Right (YZ)



Overall size X x Y x Z (mm)

125.0 x 54.0 x 24.0

Volume 57.4 cm3

Print: 46.3 g, 2.65 h (PETG, Bambu A1)

Holes (blue) dia mm / axis / qty: 5.2/z/4; 34.4/x/1

General tolerance +/-0.2 mm (FDM). Dimensions in mm.
Hidden edges not shown. See print_guide.md for orientation.

Top cover (magnetic)	Team MEDTHAI	Scale: fit	Sheet 9/9
Spiro - DIY IoT spirometer (v8) file: top_cover.stl	Drawn by Claude (auto) from CAD	Educational prototype	Rev. A